MILITARY SPECIFICATION

OVEN, THERMAL DRYING, ELECTRIC

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers one type of thermal drying electric oven (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Waterproofed Kraft
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 4430

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PUBLICATIONS

American Standards Association Standards:
C1 - National Electrical Code
(Copies may be obtained from the American Standards Association, 70 East 45th Street, New York, 18, N.Y.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction - The oven furnished under this specification shall be a product which has been inspected and has passed the preproduction examination and tests specified herein.

3.2 Design and construction - The oven shall be of the gravity convection type, equipped with thermometer, indicator light, variable thermostat control, switch or switches for controlling power input to the heaters and suitable high resistance alloy heating elements. The oven cabinet shall have a 3/8 inch insulated wall encased in a heavy steel housing and frame to conserve heat. Each oven cabinet shall be designed with self supports in the working chamber to hold shelves at various heights and two shelves shall be supplied with each oven. All electrical wiring and control circuits shall be enclosed in a wiring compartment thermally insulated from the working chamber. The construction of the oven shall permit fresh air to pass through the cabinet. Heat transfer by convection shall set up air - flow patterns for heat distribution within the working chamber conducive to accurate temperature control and the air shall be exhausted through a suitable adjustable ventilating port on top of the oven cabinet.

3.2.1. Power supply - The oven shall be designed for operation on 110 volt, 60 cycle, single phase, alternating current.

3.2.2 Operating temperature range - The operating temperature range of the oven shall be from 104°F to 392°F (40°C to 200°C) and tested within the limits of 4.6.4.

3.2.2.1 Operating Temperature - The operating temperature of the oven shall be 200°F to be reached in 20 minutes.

3.3 Material -

3.3.1 General - Materials shall conform to the applicable specifications and military standards as listed or required herein.

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Materials which are not covered by applicable specifications or standards, or which are not specifically described herein, shall be of the best quality and suitable for the purpose intended.

3.3.2 Thermal insulation - Thermal insulation used in the oven shall be high-grade, heat-insulating material which is noncombustible, nonhygroscopic, and nonpowdering.

3.3.3 Heating elements - Heating elements shall be fabricated of a high-resistance alloy suitable for use in ovens covered by this specification and capable of fulfilling the performance requirements contained herein.

3.3.4 Electrical - All electrical components, parts, insulation and circuits comprising the unit shall meet the requirements of American Standards Association Standard C1, National Electrical Code and shall be suitable for the use intended with respect to heat resistance and moisture absorption.

3.3.5 Finish - Unless otherwise specified, the welded surface of the oven shall be clean, free from dirt, scale, flux or other harmful extraneous material. The external surfaces shall have a smooth finish free from burrs, flaws, ragged and sharp edges.

3.3.5.1 Painting - Unless otherwise specified the oven shall be painted in accordance with the manufacturer's standard practice.

3.4.1 Dimensions -

3.4.1 Interior - The interior dimensions of the oven shall be 16 1/2 inches wide, 11 1/2 inches deep, 16 inches high.

3.4.2 Overall - The overall dimensions of the oven shall be 18 inches wide, 14 inches deep, 23 inches high.

3.5 Components -

3.5.1 Cabinet - The cabinet shall be of a heavy steel base and shell surrounding a 3/8 inch thick insulated wall sturdily built to give long, trouble free operation. Assembly of the cabinet shall be an all welded construction, providing space for all wiring, heaters, controls required.

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3.5.1.2 Ventilators - There shall be ventilators or openings in the walls of the oven cabinet for entrance and exit of air. The location, size, and design of the ventilators shall be consistant with good practice in the industry and the performance requirements of this specification.

3.5.1.3 Thermometer opening - A suitable opening in the top of the oven shall be provided which will hold the thermometer snugly and not permit the entrance of air into the oven.

3.5.1.4 Shelf supports - Shelf supports shall be attached to the right and left walls of the interior of the oven. The supports shall permit the shelves at different intervals from top to bottom of the working chamber, the top most position to be 3 1/2 inches from the top. Unloaded shelves shall remain horizontal when withdrawn approximately one-half the depth of the working chamber.

3.5.1.5 Thermal insulation - Thermal insulation as described in 3.3.2 shall be placed between the inner and outer walls of the cabinet. The insulation shall not support any load in excess of its own weight and shall not come in contact with uninsulated electrical conductors.

3.5.2 Doors - Oven door shall be of double-walled steel construction the same as the cabinet and shall be close fitting within the door opening to prevent heat loss. The door shall seat tightly against a suitable resilient heat resistant seal around the edge to prevent air leakage. The space between the inner and outer faces of the door shall be thermally insulated in accordance with 3.3.2 and 3.5.1.5. The door shall extend the full width and height of the working chamber and hung by a continuous hinge.

3.5.3 Shelves - Each oven shall be equipped with two removable nickel plated wire rod type steel shelves. The shelves shall be approximately the same size as the width and depth of the oven working chamber.

3.5.4 Thermometer - The oven shall be provided with a laboratory type mercury thermometer graduated from 32°F to 500°F (0°C to 260°C) in 2°F graduations. Design of the thermometer shall be such that it can be inserted through the thermometer opening in the cabinet (see 3.5.1.3) and the bulb positively positioned approximately 4 inches below the inside of the top of the oven interior and the entire range of the scale shall be visible on the stem above the top of the oven. Accuracy shall be tested within limits stated in 4.6.1.

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3.5.5 Heating units - Oven heating units provided shall be designed for use with the power supply specified in 3.2.1 and capable of producing and maintaining oven temperatures stated in 3.2.2. The heating units should be installed with means to prevent injury to personnel handling material in the oven during operation. The heating elements shall be protected against excessive dripping of liquids into either the exposed element wire or element sheath. The terminals of the heating elements shall be conveniently accessible for wiring, test and replacement purposes.

3.5.6 Switch control - Power input to the heating unit shall be controlled by a two-position switch or other input controller, which shall be separate from the thermostat control, suitable for use with the specified power supply to permit operation of the oven without an overshoot of the temperature value for which the thermostat is set, within the range given in 3.2.2. Switch positions shall be permanently and plainly marked.

3.5.6.1 Neon lamps - The oven shall be equipped with two neon lamps, one red to indicate oven "on" and the other clear for thermostat "action."

3.5.7 Thermostat - Each oven shall be equipped with an automatic hydraulic thermostat fully enclosed to protect the instrument against dust, dirt, moisture, or corrosive media. Adjustment of the thermostat shall be by means of a knob and graduated reference dial allowing for a variation in temperature within the oven as specified in 3.2.2. The thermostat sensitive element shall be suitably located inside the working chamber of the oven to insure sensitive response to temperature fluctuations and tested within the limits of 4.6.2.

3.5.8 Circuit wiring - The circuit wiring of the oven shall be completely enclosed in a control and wiring compartment integral with the cabinet. All exposed wiring shall be covered with suitably impregnated asbestos, fibrous glass sleeving or other suitable heat and moisture resisting insulation. Electrical connections shall be mechanically secure and electrically continuous and shall not be adversely affected by the heat produced as a result of normal use of the oven. Circuit wiring shall terminate at a suitable terminal box for connection to the power supply.

3.5.9 Workmanship - Workmanship shall be first class throughout and free from evidence of inferior workmanship which might affect proper functioning, appearance or impair their serviceability.

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readings shall be taken until equilibrium is reached for the thermostat setting. The range of temperatures throughout the oven and the variation from the equilibrium temperature shall not exceed the limits given in 3.2.2 and 3.2.2.1. Temperature readings shall be taken five times at five minute intervals following the time of reaching the point of equilibrium.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Levels A and B - Cleaning, preservation and packaging methods and materials shall be in accordance with MIL-P-116.

5.1.2 Level C - The equipment shall be cleaned, preserved and packaged in accordance with good commercial practice.

5.2 Packing -

5.2.1 Levels A and B - Packing shall be in accordance with MIL-P-601, (overseas type) or PPP-B-621.

5.2.2 Level C - All components of ovens shall be packed to insure carrier acceptance and safe delivery to destination. Containers shall comply with Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.3 Marking - In addition to any special marking required in the contract or order, marking of packages and containers shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The thermal drying electric oven covered by this specification are intended for curing the epoxy used for the repair of filter elements at temperatures ranging from 100°F to 165°F for a period of one hour.

6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Level of packaging and packing required (See 5.1 and 5.2)

(c) Special marking and/or data.

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(a) Thermometer accuracy test (4.6.1)

(b) Thermostat test (4.6.2)

(c) Dielectric strength test (4.6.3)

(d) Tolerance and range test (4.6.4)

4.5 Examination of product - The oven and components shall be thoroughly examined for design, construction materials and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.6 Test methods -

4.6.1 Thermometer accuracy test - Each thermometer shall be checked in a suitable bath with a National Bureau of Standards certified thermometer between 32°F and 500°F at intervals of 100°F. Thermometer under test shall be accurate to within plus or minus 1.5°F over the entire temperature range. The immersion depth of the thermometer shall be 4 inches. The test shall be conducted in a well circulated bath to insure a low temperature differential within the bath.

4.6.2 Thermostat test - The thermostat shall successfully complete 50 cycles of operation when the oven is connected to a supply circuit of current type specified in 3.2.1 having a voltage which is 120 percent of the rated voltage of the oven. The thermostat shall be so connected that there will be a difference of potential equal to the test voltage between live parts of the control mechanism and the dead metal parts of the oven. There shall be no electrical or mechanical failure of the control, nor undue burning, pitting, or welding of the contacts.

4.6.3 Dielectric strength test - A 60 cycle alternating potential of 900 volts shall be applied for a period of one minute between live parts and dead metal parts with the oven at the operating temperature reached in normal use. The test shall be conducted with a suitable transformer, the output voltage of which can be regulated. Starting at zero, the applied potential is to be increased gradually until the required test value is reached or breakdown occurs.

4.6.4 Tolerance and range test - The thermostat shall be set successively at even 50 numbered °F. At 15 minute intervals, temperature

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Naval Air Engineering Center
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-O-81397(AS) OVEN, THERMAL DRYING, ELECTRIC

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0-8878

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NOTICE OF VALIDATION

MIL-O-81397(AS)
NOTICE 1
25 August 1987

MILITARY SPECIFICATION

OVEN, THERMAL DRYING, ELECTRICAL

MIL-O-81397(AS), dated 8 Sep 1966, has been reviewed and determined to be current.

Custodian:
Navy - AS

Preparing Activity:
Navy - AS

AMSC N/A

FSC 4430

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